

IVERCEN-10 mg/ml

Solution for injection

COMPOSITION PER ML.:

Ivermectin.....	10 mg
Benzyl alcohol	10 mg
Excipients, n.q.	

PRODUCT DESCRIPTION:

Clear and colourless solution.

PHARMACODYNAMICS/PHARMACOKINETICS/ENVIRONMENTAL PROPERTIES:**Pharmacodynamic properties**

Ivermectin is an endectocide belonging to the class of macrocyclic lactones which have a single mode of action. Compounds of this class are selectively bound to chloride channels regulated by glutamate in the nerves and muscular cells of invertebrates. Consequently, cellular membrane permeability for chloride is increased with hyperpolarization of nervous and muscular cells, resulting in the death of parasite. Compounds of this class can also interact with other regulators of chloride channels, such as those governed by the neurotransmitter gamma-aminobutyric acid (GABA).

Security margin for components of this class is due to the fact that mammals do not have chloride channels regulated by glutamate; the macrocyclic lactones have low affinity for other chloride channel's regulators in mammals which do not cross easily the hematoencephalic barrier.

Pharmacokinetic properties

In each of the target species the pharmacokinetic profile following subcutaneous administration was characterised as follows (pharmacokinetic parameters presented as mean values): Following administration to cattle, C_{max} was 51 ng/ml, with a T_{max} of 43 h, T_{1/2} of 129 h and an AUC of 7398 ng.h/ml. Following two subsequent administrations seven days apart to sheep, C_{max} was 14 ng/ml, with a T_{max} of 202 h, T_{1/2} of 380 h and an AUC of 4686 ng.h/ml. Following administration to pigs, C_{max} was 6.35 ng/ml, with a T_{max} of 106 h, T_{1/2} of 219 h and an AUC of 1260 ng.h/ml. Only about 2% of the drug is excreted in urine, faecal excretion being the major route of elimination.

Tissue residues of radioactivity following subcutaneous administration of tritium-labelled ivermectin are highest in liver and fat; lowest levels are found in brain. In cattle, the residual antiparasitic effect of ivermectin is due to its persistence which in turn is due in part to its long intrinsic half life and its relatively high protein binding (90%).

Environmental Properties

EXTREMELY DANGEROUS TO FISH AND AQUATIC LIFE. Do not contaminate surface waters or ditches with product or used container. Any unused veterinary medicinal product or waste material derived from such veterinary medicinal product should be disposed of in accordance with local requirements.

TARGET SPECIES:

Cattle, sheep and pigs

INDICATIONS:

CATTLE: For the treatment of gastrointestinal nematodes, lungworms, eyeworms, warble flies, mites and lice (as shown below) of beef and non-lactating dairy cattle: Gastrointestinal worms (adults and 4th stage larvae): *Ostertagia ostertagi*, *Ostertagia lyrata*, *Haemonchus placei*, *Trichostrongylus colubriformis*, *Cooperia oncophora* (adults), *Cooperia punctata* (adults), *Cooperia pectinata* (adults), *Bunostomum phlebotomum*, *Oesophagostomum radiatum*.

Lungworms (adult and 4th stage larvae): *Dictyocaulus viviparus*.

Eyeworms (adult): *Thelazia* spp.

Warble flies (parasitic stages): *Hypoderma bovis*, *H. lineatum*. Mites: *Psoroptes ovis* *Sarcoptes scabiei* var. *bovis*.

Sucking lice: *Linognathus vituli*, *Haematopinus euryternus*, *Solenopotes capillatus*.

May also be used as an aid in the control of the mange mite *Chorioptes bovis* but complete elimination may not occur. Treatment at the recommended dose rate prevents re-infection with *Haemonchus placei*, *Cooperia oncophora*, *Cooperia pectinata* and *Trichostrongylus axei* for 7 days after treatment, *Ostertagia ostertagi* and *Oesophagostomum radiatum* for 14 days after treatment and *Dictyocaulus viviparus* for 21 days after treatment.

SHEEP: For the treatment of psoroptic mange (sheep scab), gastrointestinal nematodes, lungworms and nasal bots of sheep:

Gastrointestinal roundworms (adults): *Ostertagia circumcincta*, *Haemonchus contortus*, *Trichostrongylus axei*, *T. colubriformis* and *T. vitrinus*, *Cooperia curticei*, *Nematodirus filicollis*. Variable activity may be observed against *Cooperia curticei* and *Nematodirus filicollis*.

Lungworms: *Dictyocaulus filaria* (adults). Mange mites: *Psoroptes ovis* Nasal bot: *Oestrus ovis* (all larval stages)

PIGS: For the treatment of gastro-intestinal nematodes, lungworms, lice and mange mites of pigs.

Gastro-intestinal worms (adult and fourth stage larvae): *Ascaris suum*, *Hyostromylus rubidus*, *Oesophagostomum* spp., *Strongyloides ransomi* (adults).

Lungworms: *Metastrongylus* spp. (adults).

Lice: *Haematopinus suis*.

Mange Mites: *Sarcoptes scabiei* var. *suis*

DOSAGE AND METHOD OF ADMINISTRATION:

For single administration only (except for the treatment of *Psoroptes ovis* infections in sheep). To ensure administration of a correct dose, bodyweight should be determined as accurately as possible. Accuracy of the dosing device should be checked.

Cattle: Dosage: 200 micrograms ivermectin per kg bodyweight. **Administration:** Inject subcutaneously in front of, or behind, the shoulder using aseptic technique.

Sheep: Dosage: 200 micrograms ivermectin per kg bodyweight. **Administration:** For the treatment of gastrointestinal roundworms, lungworms and nasal bots inject once subcutaneously in the neck, using aseptic precautions. For the treatment of *Psoroptes ovis* (sheep scab), two injections with a seven day interval are required to treat clinical signs of scab and to eliminate living mites. For young lambs weighing less than 20.0 kg give 0.1 ml per 5 kg.

Pigs: Dosage: 300 micrograms ivermectin per kg bodyweight. **Administration:** The recommended route of administration is by subcutaneous injection into the neck using aseptic technique. For piglets weighing less than 16 kg give 0.1 ml per 3 kg.

CONTRAINDICATIONS:

Do not use in lactating dairy cows and sheep producing milk for human consumption. Do not use in non-lactating dairy cows, including pregnant dairy heifers or non-lactating dairy sheep within 60 days of calving/lambing. Do not use in cases of known hypersensitivity to ivermectin. Do not administer by the intravenous or intramuscular route.

WARNING AND PRECAUTION:

Care should be taken to avoid the following practices because they increase the risk of development of resistance and could ultimately result in ineffective therapy:

- Too frequent and repeated use of anthelmintics from the same class, over an extended period of time.

- Underdosing, which may be due to underestimation of bodyweight, misadministration of the product, or lack of calibration of the dosing device (if any).

Suspected clinical cases of resistance to anthelmintics should be further investigated using appropriate tests (e.g. Faecal Egg Count Reduction Test). Where the results of the test(s) strongly suggest resistance to a particular anthelmintic, an anthelmintic belonging to another pharmacological class and having a different mode of action should be used. Treatment of psoroptic mange (sheep scab) with one injection is not recommended because, although clinical improvement may be seen, elimination of all mites may not occur. Sheep scab (*Psoroptes ovis*) is an extremely contagious external parasite of sheep. Following treatment of infected sheep great care must be taken to avoid re-infestation as mites may be viable for up to 15 days out of the sheep. It is important to ensure all sheep which have been in contact with infected sheep are treated. Contact between treated infected sheep and non-treated, non-infected flocks must be avoided until at least 7 days after the last treatment. Resistance to ivermectin has been reported in *Ostertagia circumcincta* in lambs and in *Ostertagia ostertagi* in cattle. Therefore, the use of this product should be based on local (regional, farm) epidemiological information about susceptibility of these helminth species and recommendations on how to limit further selection for resistance to anthelmintics.

SPECIAL PRECAUTIONS FOR USE:

i) Special precautions for use in animals

Avermectins may not be well tolerated in all non-target species. Cases of intolerance with fatal results are reported in dogs - especially Collies, Old English Sheepdogs and related breeds and crosses, and also in turtles/tortoises.

Do not combine treatment with vaccination against lungworms. If vaccinated animals are to be treated, treatment should not be carried out within a period of 28 days before or after vaccination. The shedding of nematode eggs can continue for some time after treatment.

In cattle: To avoid secondary reactions due to the death of *Hypoderma larvae* in the oesophagus or in the spine, it is recommended to administer the product at the end of warble fly activity and before the larvae reach their resting sites. Swab septum before removing each dose.

ii) Special precautions to be taken by the person administering the veterinary medicinal product to animals

Do not smoke, eat or drink while handling the product. Wash hands after use. Take care to avoid self-injection: the product may cause local irritation and/or pain at the site of injection.

INTERACTIONS WITH OTHER MEDICINES:

Do not combine ivermectin treatment with vaccination against lungworms. If vaccinated animals are to be treated, treatment should not be carried out within a period of 28 days before or after vaccination.

USE DURING PREGNANCY AND LACTATION:

Do not use in lactating dairy cows and sheep producing milk for human consumption. Do not use in non-lactating dairy cows, including pregnant dairy heifers or non-lactating dairy sheep within 60 days of calving/lambing.

ADVERSE EFFECTS/UNDESIRABLE EFFECTS:

In a few cases, clinical sign of pain and/or inflammation reaction at the injection site have been observed. These reactions are transient and disappear within treatment. Transient pain has been observed in very rare cases.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Clinical symptoms of ivermectin toxicity include ataxia and depression. No antidote has been identified. In case of overdose, symptomatic treatment should be given. No signs of toxicity were observed in animals treated at up to 3 times the recommended dose rate.

INCOMPATIBILITIES:

In the absence of compatibility studies, this veterinary medicinal product must not be mixed with other veterinary medicinal products.

WITHDRAWAL PERIOD(S):

CATTLE: Meat and offal: 49 days. Do not use in lactating cows producing milk for human consumption. Do not use in non-lactating dairy cows, including pregnant dairy heifers, within 60 days of calving.

SHEEP: Meat and offal: 42 days. Do not use in lactating ewes producing milk for human consumption. Do not use in sheep which are intended to produce milk for human consumption within 60 days of lambing.

PIGS: Meat and offal: 28 days.

SHELF-LIFE:

Shelf-life: 2 years from the manufacturing date.

Shelf-life after first opening of the container: 28 days.

STORAGE CONDITIONS:

Store below 30°C in a dry place, protected from light.

PACK SIZES:

Vials containing 50 ml., 100 ml., 250 ml. and 500 ml.

DATE OF REVISION OF PACKAGE INSERT:

03/06/2020

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